GSE 518
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Suggested Solutions to Problem Set 8

1.
(a) We know that the mean and variance of a $Gamma(\alpha_0, \beta_0)$ distribution are $E(X) = \alpha_0\beta_0$ and $Var(X) = \alpha_0\beta_0^2$. The sample moments for the mean and variance are $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ and $\widetilde{S}^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2$. The method of moment estimators solve the system of equations:

$$\begin{aligned} E(X) &= \alpha\beta = \overline{X} \\ Var(X) &= \alpha\beta^2 = \widetilde{S}^2 \end{aligned}$$

Solving for α and β we find: $\widehat{\alpha}_{MOM} = \frac{\overline{X}^2}{\widetilde{S}^2}$ and $\widehat{\beta}_{MOM} = \frac{\widetilde{S}^2}{\overline{X}}$.
(b) If we calculate the mean and variance of this density we find: $E(X) = \theta_{01}$ and $Var(X) = \theta_{02}^2$. The sample moments for the mean and variance are the same as in part (a). The method of moment estimators solve the system of equations:

$$\begin{aligned} E(X) &= \theta_1 = \overline{X} \\ Var(X) &= \theta_2^2 = \widetilde{S}^2 \end{aligned}$$

Solving for θ_1 and θ_1 we find: $\widehat{\theta}_{1MOM} = \overline{X}$ and $\widehat{\theta}_{2MOM} = \sqrt{\widetilde{S^2}} = \widetilde{S}$.

2.
(a) In this case the log of the pmf is given by

$$\ln f(x|\theta) = x\ln(\theta) - \theta - \ln(x!)$$

and therefore the log-likelihood function is given by

$$L(\theta|X_1, ..., X_n) = \sum_{i=1}^{n} \ln f(X_i|\theta) = \sum_{i=1}^{n} X_i \ln(\theta) - n\theta - \sum_{i=1}^{n} \ln(X_i!)$$

Maximizing this function with respect to θ we find

$$\frac{\partial L(\theta|X_1, ..., X_n)}{\partial\theta} = \frac{1}{\theta}\sum_{i=1}^{n} X_i - n = 0$$

and therefore the MLE for θ_0 is

$$\widehat{\theta}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}$$

Note that in principle we would have to check the second order sufficient conditions for a maximum. However, for simplicity we will not do it in the answer key (also, in the test you can feel free not to check the second order conditions).

(b) In this case the log of the pdf is given by

$$\ln f(x|\theta) = -\ln(\theta) - \frac{1}{\theta}x$$

and therefore the log-likelihood function is given by

$$L(\theta|X_1, ..., X_n) = -n\ln(\theta) - \frac{1}{\theta}\sum_{i=1}^{n} X_i$$

Maximizing this function with respect to θ we find

$$\frac{\partial L(\theta|X_1, ..., X_n)}{\partial \theta} = -\frac{n}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^{n} X_i = 0$$

Solving for θ we find the MLE for θ_0 is

$$\widehat{\theta}_{MLE} = \frac{1}{n}\sum_{i=1}^{n} X_i = \overline{X}$$

(c) In this case the log of the pdf is given by

$$\ln f(x|\theta) = \ln\left(\frac{1}{2}\right) - |x - \theta|$$

and therefore the log-likelihood function is given by

$$L(\theta|X_1, ..., X_n) = \sum_{i=1}^{n} \ln f(X_i|\theta) = n\ln\left(\frac{1}{2}\right) - \sum_{i=1}^{n} |X_i - \theta|$$

As usual, we need to maximize this function with respect to θ. However, note that in this case we cannot use the direct "calculus" approach we used in parts (a) and (b) because the absolute value function is nondifferentiable.

Maximizing the log-likelihood in this case is equivalent to minimizing the average absolute deviations of X_i from θ (or the average absolute error), $\frac{1}{n}\sum_{i=1}^{n} |X_i - \theta|$. It can be shown that the MLE for θ_0 in this case is the sample median of X_i, or $\widehat{\theta}_{MLE} = samplemedian\{x_i\}$. For a formal derivation of this result using order statistics (which we did not cover in class), see the attached paper.

3.

(a) Given this density, the expected value of X is equal to $E(X) = \int_0^1 x\theta_0 x^{\theta_0 - 1}dx = \frac{\theta_0}{\theta_0 + 1}$. If we set $E(X) = \frac{\theta}{\theta+1} = \overline{X}$ and solve for θ we find $\widehat{\theta}_{MOM} = \frac{\overline{X}}{1-\overline{X}}$.

Now we calculate the MLE. In this case the log of the pdf is given by

$$\ln f(x|\theta) = \ln(\theta) + (\theta - 1)\ln(x)$$

and therefore the log-likelihood function is given by

$$L(\theta|X_1, ..., X_n) = n\ln(\theta) + (\theta - 1)\sum_{i=1}^{n} \ln(X_i)$$

maximizing this function with respect to θ we find

$$\frac{\partial L(\theta|X_1, ..., X_n)}{\partial \theta} = \frac{n}{\theta} + \sum_{i=1}^{n} \ln(X_i) = 0$$

Solving for θ we find the MLE for θ_0 is

$$\widehat{\theta}_{MLE} = \frac{-n}{\sum_{i=1}^{n} \ln(X_i)}$$

This is an example where $\widehat{\theta}_{MLE} \neq \widehat{\theta}_{MOM}$.
(b) We know the mean of a $Uniform(0, \theta_0)$ is $E(X) = \theta_0/2$. Setting $E(X) = \theta/2 = \overline{X}$ and solving for θ we find $\widehat{\theta}_{MOM} = 2\overline{X}$.
Now we calculate the MLE (*Note: this part is only for general knowledge*). Using the hint given in the problem set, we can write the *pdf* of X as $f(X_i|\theta_0) = \frac{1}{\theta_0} \cdot 1\{X_i \leq \theta_0\}$, and hence the likelihood function is given by

$$\mathcal{L}(\theta|X_1, ..., X_n) = \prod_{i=1}^{n} f(X_i|\theta) = \frac{1}{\theta^n} \prod_{i=1}^{n} 1\{X_i \leq \theta\}$$

and the log-likelihood function by[1]

$$L(\theta|X_1, ..., X_n) = -n\ln(\theta) + \sum_{i=1}^{n} \ln(1\{X_i \leq \theta\})$$

In this case we cannot use calculus to find the maximum of this function with respect to θ because the support of X (and hence indicator function) depends on θ. On the other hand, we still need to find the maximum of the log-likelihood with respect to θ to obtain the MLE for θ_0.
Note that if we choose θ to be less than any of the X_i, then $L(\cdot) \to -\infty$, as the natural logarithm is not defined at zero. Therefore, we need to have $\theta \geq \max\{X_i\}$. Once we impose the restriction that $\theta \geq \max\{X_i\}$, maximizing the log-likelihood is equivalent to maximizing $-n\ln(\theta)$ subject to $\theta \geq \max\{X_i\}$. The function $-n\ln(\theta)$ is strictly decreasing in θ, so we want to choose the value of θ to be as small as possible subject to $\theta \geq \max\{X_i\}$, and hence, we have that $\widehat{\theta}_{MLE} = \max\{X_i\}$.

[1] Alternatively, by noting that $\prod_{i=1}^{n} 1\{X_i \leq \theta\} = 1\{\max\{X_i\} \leq \theta\}$, we can write the log-likelihood as $L(\theta|X_1, ..., X_n) = -n\ln(\theta) + \ln(1\{\max\{X_i\} \leq \theta\})$.

This is another example where $\widehat{\theta}_{MLE} \neq \widehat{\theta}_{MOM}$. In this case note that it is much easier to work with $\widehat{\theta}_{MOM}$ than with $\widehat{\theta}_{MLE}$ (for instance, to derive the mean, variance and asymptotic properties of the estimator).

4.

(a) In this case the log of the pdf is given by

$$\ln f\left(x|\theta\right) = -\frac{1}{2}\ln\left(2\pi\right) - \frac{1}{2}\ln\left(\sigma^2\right) - \frac{1}{2\sigma^2}x^2 \tag{1}$$

and therefore the log-likelihood function is given by

$$L\left(\theta|X_1,...,X_n\right) = -\frac{n}{2}\ln\left(2\pi\right) - \frac{n}{2}\ln\left(\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}X_i^2$$

maximizing this function with respect to θ we find

$$\frac{\partial L\left(\theta|X_1,...,X_n\right)}{\partial\theta} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}X_i^2 = 0$$

Solving for σ^2 we find the MLE for σ_0^2 is

$$\widehat{\sigma}_{MLE}^2 = \frac{1}{n}\sum_{i=1}^{n}X_i^2$$

NOTE: When working with σ_0^2 one has to be careful that the parameter is σ_0^2 and NOT σ_0. Hence, all the derivatives are taken with respect to σ_0^2, and not σ_0.

(b) By definition the score function is given by $S\left(X,\theta\right) = \frac{\partial \ln f(x,\theta)}{\partial\theta}$. In this case, using (1) we have

$$S\left(X,\sigma^2\right) = \frac{\partial \ln f\left(x,\sigma^2\right)}{\partial\sigma^2} = -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4}X^2 \tag{2}$$

(c) the score identity states that $E\left[S\left(X,\sigma_0^2\right)\right] = 0$. Note that $S\left(X,\sigma^2\right)$ is now evaluated at $\sigma^2 = \sigma_0^2$. Here we have

$$\begin{aligned} E\left[S\left(X,\sigma_0^2\right)\right] &= E\left[-\frac{1}{2\sigma_0^2} + \frac{1}{2\sigma_0^4}X^2\right] \\ &= -\frac{1}{2\sigma_0^2} + \frac{1}{2\sigma_0^4}E\left[X^2\right] \\ &= -\frac{1}{2\sigma_0^2} + \frac{\sigma_0^2}{2\sigma_0^4} = 0 \end{aligned}$$

where we used in the third line the fact that $E\left[X^2\right] = Var\left(X\right) = \sigma_0^2$ (since $E\left(X\right) = 0$).

(d) The information matrix identity states that $E\left[S^2\left(X,\sigma_0^2\right)\right] = -E\left[\frac{\partial^2 \ln f\left(x,\sigma_0^2\right)}{\partial(\sigma^2)^2}\right]$. In this case we have

$$\begin{aligned} E\left[S^2\left(X,\sigma_0^2\right)\right] &= E\left[\left(-\frac{1}{2\sigma_0^2}+\frac{1}{2\sigma_0^4}X^2\right)^2\right] \\ &= E\left[\frac{1}{4\sigma_0^4}-\frac{X^2}{2\sigma_0^6}+\frac{1}{4\sigma_0^8}X^4\right] \\ &= \frac{1}{4\sigma_0^4}-\frac{E\left[X^2\right]}{2\sigma_0^6}+\frac{1}{4\sigma_0^8}E\left[X^4\right] \\ &= \frac{1}{4\sigma_0^4}-\frac{\sigma_0^2}{2\sigma_0^6}+\frac{3\sigma_0^4}{4\sigma_0^8} \\ &= \frac{1}{2\sigma_0^4} \end{aligned}$$

where in the fourth line we used the fact that $E\left(X^2\right) = Var\left(X\right) = \sigma_0^2$ and $E\left(X^4\right) = 3\sigma_0^4$ (Note: this is true for a normal distribution. You can easily verify it using the moment generating function with $\mu = 0$).
Using (2) we can calculate $-E\left[\frac{\partial^2 \ln f\left(x,\sigma_0^2\right)}{\partial(\sigma^2)^2}\right]$ as

$$\begin{aligned} -E\left[\frac{\partial^2 \ln f\left(x,\sigma_0^2\right)}{\partial(\sigma^2)^2}\right] &= -E\left[\frac{\partial S\left(x,\sigma_0^2\right)}{\partial\sigma^2}\right] \\ &= -E\left[\frac{1}{2\sigma_0^4}-\frac{1}{\left(\sigma_0^2\right)^3}X^2\right] \\ &= -\frac{1}{2\sigma_0^4}+\frac{1}{\left(\sigma_0^2\right)^3}E\left[X^2\right] \\ &= -\frac{1}{2\sigma_0^4}+\frac{1}{\left(\sigma_0^2\right)^3}\sigma_0^2 \\ &= \frac{1}{2\sigma_0^4} \end{aligned}$$

Hence, the information matrix identity holds.
(e) In this case the log of the pdf is given by

$$\ln f\left(x|\sigma^2\right) = -\frac{1}{2}\ln\left(2\pi\right)-\frac{1}{2}\ln\left(\sigma^2\right)-\frac{X^2}{2\sigma^2}$$

The expected value of $\ln f\left(x|\theta\right)$ taking into account the fact that X has a distribution $N\left(0,\sigma_0^2\right)$ is:

$$\begin{aligned} E\left[\ln f\left(x|\sigma^2\right)\right] &= -\frac{1}{2}\ln\left(2\pi\right)-\frac{1}{2}\ln\left(\sigma^2\right)-\frac{E\left[X^2\right]}{2\sigma^2} \\ &= -\frac{1}{2}\ln\left(2\pi\right)-\frac{1}{2}\ln\left(\sigma^2\right)-\frac{\sigma_0^2}{2\sigma^2} \end{aligned}$$

The first order condition for maximizing this function with respect to σ^2 is

$$\frac{\partial L\left(\sigma^2|X_1,...,X_n\right)}{\partial\sigma^2}=-\frac{1}{2\sigma^2}+\frac{\sigma_0^2}{2\left(\sigma^2\right)^2}=0$$

Solving for σ^2 yields that the value of σ^2 that maximizes $E\left[\ln f\left(x|\theta\right)\right]$ is σ_0^2. You can also check that the second order condition for a maximum holds in this case.

5. Since in this case we want to estimate five parameters, we need five moments of (X,Y). We use the first two moments plus $E\left(XY\right)$ and set them equal to their sample analogues. Hence, the five equations to use are given by

$$\begin{aligned}
E\left(X\right) &= \mu_x=\overline{X}\\
E\left(Y\right) &= \mu_y=\overline{Y}\\
Var\left(X\right) &= \sigma_x^2=\frac{1}{n}\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2\\
Var\left(Y\right) &= \sigma_y^2=\frac{1}{n}\sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2\\
E\left(XY\right) &= \mu_x\mu_y+\rho\sigma_x\sigma_y=\frac{1}{n}\sum_{i=1}^{n}X_iY_i
\end{aligned}$$

The MOM estimators are then given by:

$$\begin{aligned}
\widehat{\mu}_x &= \overline{X}\\
\widehat{\mu}_y &= \overline{Y}\\
\widehat{\sigma}_x^2 &= \frac{1}{n}\sum_{i=1}^{n}\left(X_i-\overline{X}\right)^2\\
\widehat{\sigma}_y^2 &= \frac{1}{n}\sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2\\
\widehat{\rho} &= \frac{\frac{1}{n}\sum_{i=1}^{n}X_iY_i-\overline{X}\,\overline{Y}}{\widehat{\sigma}_x\widehat{\sigma}_y}
\end{aligned}$$



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$$\bar{X}_{n+1} = (n\bar{X}_n + X_{n+1})/(n+1) \quad (1)$$

$$ns_{n+1}^2 = (n-1)s_n^2 + \frac{n}{(n+1)}(X_{n+1} - \bar{X}_n)^2. \quad (2)$$

(See below for details.) Now, $\bar{X}_n$ and X_{n+1} are independent, $N(\mu, \sigma^2/n)$ (by the induction hypothesis) and $N(\mu, \sigma^2)$, respectively. Hence $\bar{X}_{n+1}$ is a linear combination of two independent normal random variables, and (a) follows by simply computing $E\bar{X}_{n+1}$ and $V(\bar{X}_{n+1})$. Similarly, it follows that $X_{n+1} - \bar{X}_n$ has a $N(0, ((n+1)/n)\sigma^2)$ distribution, and so $(n/(n+1))(X_{n+1} - \bar{X}_n)^2$ is distributed as the square of a $N(0, \sigma^2)$ random variable. Since X_{n+1} is independent of s_n^2, and $\bar{X}_n$ is also independent of s_n^2 by the induction hypothesis, (b) follows after dividing (2) through by σ^2. Finally, the induction hypothesis (and inspection of (1)) shows that $\bar{X}_{n+1}$ is independent of s_n^2, and (c) follows by noting

$$\operatorname{cov}(n\bar{X}_n + X_{n+1}, X_{n+1} - \bar{X}_n) = \sigma^2 - n \cdot \sigma^2/n = 0.$$

The relationships (1) and (2) are themselves nice exercises in summation notation. (1) is direct, as is the useful consequence $\bar{X}_{n+1} - \bar{X}_n = (X_{n+1} - \bar{X}_n)/(n+1)$. Formula (2) follows by expanding

$$\sum_{i=1}^{n+1}(X_i - \bar{X}_{n+1})^2 = \sum_{i=1}^{n+1}[(X_i - \bar{X}_n) + (\bar{X}_n - \bar{X}_{n+1})]^2$$

$$= \sum_{i=1}^{n+1}(X_i - \bar{X}_n)^2 + 2(\bar{X}_n - \bar{X}_{n+1}) \times \sum_{i=1}^{n+1}(X_i - \bar{X}_n) + (n+1)(\bar{X}_n - \bar{X}_{n+1})^2.$$

Then, noting that

$$\sum_{i=1}^{n+1}(X_i - \bar{X}_n)^2 = (n-1)s_n^2 + (X_{n+1} - \bar{X}_n)^2,$$

and

$$\sum_{i=1}^{n+1}(X_i - \bar{X}_n) = \sum_{i=1}^{n}(X_i - \bar{X}_n) + (X_{n+1} - \bar{X}_n) = X_{n+1} - \bar{X}_n,$$

(2) follows readily.

One feature of this proof is that it only involves bivariate distributions; hence the independence of $X_1 + X_2$ and $X_1 - X_2$ and of $n\bar{X}_n + X_{n+1}$ and $X_{n+1} - \bar{X}_n$ can be illustrated geometrically at a blackboard. In addition, formulas (1) and (2) are of independent interest as useful algorithmic devices for calculating $\bar{X}$ and s^2 on a computer; see Chan, Golub, and LeVeque (1983) for a discussion and evaluation of these and other algorithms.

Kruskal's original version of this proof did not assume the knowledge that a lack of correlation among bivariate normal random variables implies independence; instead he derived the relevant distributions analytically by transformation.

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The Double Exponential Distribution: Using Calculus to Find a Maximum Likelihood Estimator

ROBERT M. NORTON*

1. INTRODUCTION

When learning how to derive a maximum likelihood estimator (MLE) for an unknown parameter of a known density function, students are tempted to stop after equating the first derivative to zero and solving, particularly if the derivative is of complicated form. Yet students know from calculus that a function may have a maximum, minimum, or neither at a critical number. The purpose of this note is to give a simple maximization argument on one particular exercise encountered by any teacher using the mathematical statistics text by Hogg and Craig. It is an exercise students always find difficult.

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2. THE PROBLEM AND SOLUTION

The particular problem is this: Find the MLE for θ when $X_1, X_2, \ldots, X_n$ represents a random sample from the density

$$f(x; \theta) = \tfrac{1}{2}\exp(-|x - \theta|), \quad -\infty < x < \infty, \quad -\infty < \theta < \infty.$$

The likelihood function $L(\theta)$ has as its natural logarithm

$$\psi(\theta) = -n \ln 2 - \sum_{i=1}^{n} |x_i - \theta|$$

$$= -n \ln 2 - \sum_{i=1}^{n} \{(x_i - \theta)^2\}^{1/2}.$$

The function ψ is continuous everywhere and differentiable except at $\theta = x_1, x_2, \ldots, x_n$. Now when it exists,

$$\psi'(\theta) = \sum_{i=1}^{n} \{(x_i - \theta)^2\}^{-1/2} (x_i - \theta)$$

$$= \sum_{i=1}^{n} (x_i - \theta)/|x_i - \theta|,$$

a sum of ones and minus ones.

Let $y_1 < y_2 < \cdots < y_n$ denote the order statistics of the sample. For $\theta < y_1$, $\psi'(\theta) = n$ since $(x_i - \theta)/|x_i - \theta| = 1$ for all i. If $y_1 < \theta < y_2$ then $\psi'(\theta) = n - 2$, while if $y_2 < \theta < y_3$ then $\psi'(\theta) = n - 4$, and so forth. The graph of ψ is a continuous polygonal curve. It is now easy to see that for n odd, ψ is strictly increasing on $(-\infty, y_{(n+1)/2}]$ and strictly decreasing on $[y_{(n+1)/2}, \infty)$, so that at $\theta = y_{(n+1)/2}$ the functions ψ and L are maximized. If n is even, the highest points on the graph of L lie on the horizontal segment

$$\{(\theta, L(\theta)) : y_{n/2} \leq \theta \leq y_{(n/2)+1}\}.$$

Hence the sample median $\tilde{x}$ is a MLE for θ. My students have found this argument helpful on this problem.

Students who realize that the derivative is a sum of ones and minus ones sometimes try to equate the derivative to zero, reasoning that there must be as many ones as minus ones, and conclude that the median is an answer. Yet this makes sense only for n even, and then it is known only that $\tilde{x}$ is a critical number for θ. Still more work is required to show that L is maximized at $\theta = \tilde{x}$.

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